Blockchain and Green Certificates Based Market Structure and Transaction Mechanism of Direct Power-Purchase for Industrial Users

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***Abstract*—Nowadays, Direct Power Purchase (DPP) for industrial users has become an important part of the electricity market. This type of transaction usually needs to rely on regulatory agencies to assist in the transaction, and the transaction algorithm is relatively simple. Blockchain technology based on encryption algorithm has the function of distributed ledger, which can make DPP more transparent and reliable. This paper proposes a DPP transaction method, which incorporates the peak shaving characteristics of industrial users and the green certificate system into transactions, and completes related transactions on the blockchain platform. First, establish a DPP market structure based on blockchain technology. Second, establish a DPP transaction mechanism based on green certificates and peak shaving characteristics. Simulation results for four cases show that the proposed method can respond to peak shaving capacity and bring more profit to renewable energy companies.**

***Index Terms*—Blockchain, direct power-purchase, green certificate, peak shaving, quota system.**

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NOMENCLATURE

A. Abbreviations

DPP	Direct power-purchase
FDI	Fake data injection
CA	Certificate authority

B. Sets and Indices

m	Set of electricity sellers
n	Set of industrial users
i	Index for electricity sellers
j	Index for industrial users

C. Parameters

$Q_{B,j}$	Total electricity purchased by industrial user *j* (MWh)
$Q_{S,i}$	Total electricity sold by electricity seller *i* (MWh)
S_{ij}	Subsidies for Direct Power-Purchase by industrial users (Token)
N_{gr}	Number of green certificates traded
P_{gr}	Price of green certificates (Token/certificate)
σ	Price adjustment ratio
$\Delta\beta$	Transmission loss rate between regions
τ	Transmission costs between regions
ΔW	Environmental subsidies
L_x	Comprehensive adjustment load of industrial user *x* (MWh)
L_α	Start-stop peak shaving power (MWh)
L_β	Load shifting power (MWh)
L_γ	Start-stop peak shaving power (MWh)
w_α	Weighting factor for start-stop loads
w_β	Weighting factor for load shifting
w_γ	Weighting factor for increasable loads

D. Variables

W	Social welfare of the market
$P_{B,j}$	Power purchase bid of industrial user *j* (Token/kWh)
$P_{S,i}$	Selling price of electricity seller *i* (Token/kWh)
Q_{ij}	The amount of electricity traded between electricity seller i and industrial user *j* (MWh)
M_{ij}	Matching element consisting of *i* and *j*
P_j^i	The final electricity purchase price between *i* and *j* (Token/kWh)
P_i^j	The final electricity sale price between *i* and *j* (Token/kWh)

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I. Introduction

With the diversification of electricity maket entities, power trading methods are also enriched, in which direct power purchase (DPP) is one of the most important components. DPP refers to a transaction form in which enterprises or individuals with a certain power consumption scale directly purchase electricity from generation units [1]. This kind of transaction form introduces competition on the power sales side and improves the optimal allocation level [2].

However, there are loopholes in the credibility and security of DPP. Since there is no third-party operator in DPP, and the transaction lacks incentives and transparency, both parties have the risk of transaction failure [3]. In addition, only buyers and sellers with a two-way communication flow are involved in the transaction process, which is an efficient but simple mode. The bidirectional information flow can pose serious challenges to the security and privacy of consumption and transaction data [4]. As mentioned in reference [5], the power system may be attacked by fake data injection (FDI), which hinders the operation and trading of the electricity market.

To address those issues of trust and transparency in the peer-to-peer transaction form of DPP, blockchain, as a distributed ledger system, is introduced to be combined with the DPP transaction process. Blockchain technology has the characteristics of decentralization and traceability [6], its data transaction authentication method based on the federated chain and smart contract solves the data security problem and realizes the auditability, accountability, and integrity of DPP [7], [8], [9]. Due to its decentralization and transparency, blockchain can be deeply integrated with grid management [10], price mechanism [11], electricity market transactions [12], and power trading [13]. For example, blockchain can be applied to power system management by integrating smart contracts, and it can also be applied to P2P energy trading platforms according to the characteristics of its distributed information [14]. A large number of references have studied the extensive application of blockchain as a power trading platform for its implementation effect and efficiency [15], [16]. Blockchain can formulate EV charging distribution schemes using decentralized smart contracts and ensure safe peer-to-peer energy transactions between producers and consumers [17], [18]. Contract theory and incentive mechanisms are used to construct a blockchain-based energy transaction model and framework to ensure the safety and efficiency of transactions [19]. Reference [20] adopts the blockchain for medium and long-term energy trading and proved that this method is helpful to promote the grid-connected power generation of renewable energy and improve social benefits. To sum up, the blockchain technology has five main aspects of the benefits on the DPP: 1) Enhanced security; 2) Greater transparency; 3) Instant traceability; 4) Increased efficiency and speed; 5) Automation.

Aiming to control the greenhouse gas emissions, multiple policies on carbon emissions trading has been issued [21] and renewable energy like wind and solar are in an explosive development stage. Green certificate, a tradable asset that proves that electricity has been generated by a renewable energy source, is one of the carbon-limitation policies. It can be traded in a similar manner to the DPP, mainly in the form of a listing [22]. The government would impose a cost on non-renewable production for the generation side of DPP and set a maximum emission limitation to the total emissions, and therefore, green certificates create extra revenue for green production. However, previous studies on DPP [14], [15], [16], [17], [18], [19], [20] did not take the green certificate transaction into consideration. With the increasing proportion of renewable energy, the impact of green certificate transactions on electricity trading becomes non-negligible. Therefore, the application of green certificate mechanisms should be fully considered in the blockchain-based DPP, which can help promote the optimal allocation of renewable energy [23].

Moreover, the electricity market transaction is closely related to the stable operation of the power system. The market could realize peak-shaving and valley-filling through price signals, and the choice of peak shaving solution will in turn influence the cost of electricity procurement. For the continuous expansion of the peak-to-valley difference faced by the power system, it can be alleviated by giving subsidies or adjusting the electricity price during peaks and valleys. For renewable energy generation and controllable energy storage equipment, peak shaving can be performed according to their power demand curve, thereby minimizing electricity costs [24]. The impact of high penetration variable renewable energy generation on electricity prices has been studied in reference [25]. In addition, previous studies [26] and [27] consider the flexible peak shaving resource cost in carbon trading and establish a collaborative optimization model for safety and environmental protection with the lowest peak shaving cost target. As the DPP becomes a development trend in current electricity market, considering the peak regulation effect in DPP will have positive impacts on the power system operation.

However, few studies have combined the above-mentioned two aspects in the DPP transaction of industrial customers. Therefore, based on our previous work [28], this paper is oriented to the industrial users and proposes an architecture and method of blockchain-based DPP for them. It should be noted that this paper mainly focuses on the DPP issues so that the peak shaving and valley filling issues and the market regulatory framework are out of the scope of the paper. The contributions of this paper could be summarized as follows:

1) A blockchain-based DPP transaction architecture considering the green certificate system and the peak shaving capacity of the industrial users is proposed, which increases the consumption of renewable energy, ensures the power system with sufficient peak shaving capacity and brings more profits to renewable energy generators.
2) Two different kinds of smart contracts (i.e., DPP smart contract and peak shaving smart contract) are designed, which is the most essential components of the DPP transaction blockchain and ensure the transactions program running and executing automatically.
3) The feasibility and effectiveness of the proposed DPP model are verified through the comparison with the other three benchmark methods based on the Z-ledger platform.

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The rest of the paper is organized as follows: Section II describes the blockchain encryption algorithm and the design of the smart contract for the DPP problem. Section III focuses on the methodology and the rationale analysis of the combination of blockchain and DPP transactions. Section IV presents the case study and analysis. Section V highlights the conclusions and future work.

II. Materials on Blockchain and Power Transaction

A. Blockchain Encryption Algorithm

Blockchain is a technology for decentralized, distributed data accounting and storage. In contrast to conventional centralized and distributed counting and storing technologies, blockchain counts on decentralized ledgers stored in multi-nodes in the network. Each node contains a full set of ledgers that captures node information and on-chain data to assure the reliability of each node's data.

The structure of a blockchain can be divided into a block header and a block body. The block header consists of a timestamp and a nonce. It contains the Merkle root nodes, including the user side, the generation side, and the transaction center side, in addition to recording the hash value of the preceding block. Each chain records the hash values of the child nodes in a binary tree structure. The block body is the part that contains the block header and the child nodes of the Merkle tree. The distributive database has several attribute records, which can store the transaction information or some other information. Blockchains are categorized as public, private, or consortium chains depending on their openness. The consortium chain's organizational structure and authority management are most appropriate for the transaction scenario of the electricity market. Based on the standard structure of the consortium chain, market players need to be divided into nodes on the basis of actual business requirements and functions. In the DPP market devised in this paper, the main subject is classified into two types of nodes, namely, transaction nodes and supervision nodes. Among them, the supervisory nodes are the main nodes with the highest authority to access to the database of the whole system, and they are the managers of accounting. Market entities such as industrial users and power generation companies are transaction nodes that only have transaction privileges while without accounting or access privileges. The main functions of the transaction nodes are: sending transaction messages, receiving transaction information, and accounting for the content of transactions throughout the network nodes.

The hash algorithm is a mathematical one-way function that garbles data and makes it unreadable. The asymmetric encryption technology makes up the bulk of the traditional blockchain encryption technique. The main method is to calculate the hash value of the current block by applying a hashing algorithm in the block. After that, asymmetric encryption algorithms are applied to transfer transactions. Its operating principle is shown in Fig. 1.

A public key and a private key are assigned to each participant in the DPP market. The private key is used to digitally sign the transaction content as it is written by the trading entity. The recipient user receives both the transaction content and the signature based on the blockchain network. By explaining the digital signature, the communicated material and comparing the digest content, the receiving users will verify the data.

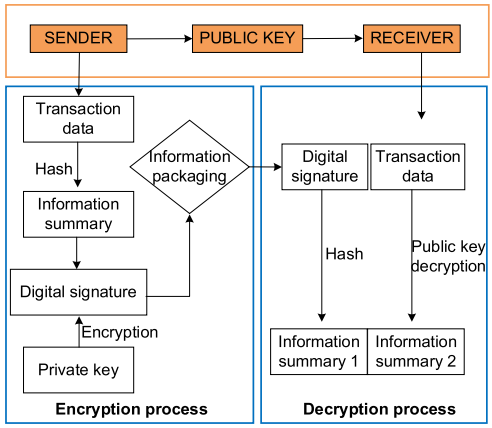


Fig. 1. Blockchain encryption process.

B. Design of Smart Contract for DPP

Smart contracts are a sort of Turing-complete contractual software. At the time of its founding, the initial version of Bitcoin served as a representation of the blockchain because it had just monetary qualities and no contractual capabilities. With the development of blockchain technology, the second generation of smart contracts in blockchain Ethereum has grown into chain codes in the third generation of blockchain hyperledger. The chain code defines an asset or a set of assets that can define or resolve multiple related contracts. Power generation companies and industrial users among DPP market entities can make tranction through the DPP chain code. Consensus techniques are mostly used to standardize the underlying protocols that underpin the functioning of contracts in chain code. The core elements of the conventional consensus mechanism include node election, log replication and security guarantees. The three types of internal nodes are leaders, followers, and electors. The smart contracts are essential to the integration of the electricity market and blockchain. As an automatic execution tool, it can effectively prevent the content of the transaction from being tampered with and also prevent property loss due to post-transaction default.

When designing a DPP contract based on the quota system, the main elements are stated as:

- At the time of market registration, the listing contract is triggered, and each participant supplies trading electricity and quotations. Meanwhile, a portion of the deposit is made in the capital account so that the pending order can be recorded after the payment has been made;
- Once the trade listing cycle is complete, the transaction regulator will perform a security check, execute the check contract, and upload the check order to the matching system. The contract will issue a warning and instruct both parties to relist for unsatisfactory orders;
- When the transaction enters the settlement stage, the clearing contract is triggered in accordance with the suggested

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matching process, and the transaction amount and volume are automatically determined without operator involvement. The security and reliability of power transactions are guaranteed.

When designing the smart contract for the peak shaving, this procedure can be described as:

- When the quota allocation function is invoked, the regulatory agency assigns quotas to subjects based on indicators and automatically executes the chain code. Each transaction entity needs to pay a certain amount of liquidated damages to the contract. The trading model based on a market mechanism can effectively motivate businesses to reduce emissions when compared to the traditional model of government-organized emissions trading;
- The chain code automatic decision-making system will combine with the Internet of Things technology to collect the execution indications of each market participant and automatically apply the matching algorithm when a peak-shaving allocation performance cycle is finished. It avoids the problem of participants in some user systems evading the responsibility of peak shaving and can give full play to the incentive effect of emission reduction awards and penalties;
- When a corporation enters the transaction settlement stage, it automatically performs transaction settlement operations. Chaincode-based transactions can increase the effectiveness of chipping allocation and reduce transaction costs because they do not require manual verification by the appropriate departments.

According to the trading process, as soon as the DPP market and the peak-shaving contracts are deployed on the chain, they will run in parallel and execute the transactions automatically according to the program. On the one hand, the content of the transactions on the chain can be used as transaction credentials. On the other hand, it can be used as a distributed database for accessing to the transaction ledger later.

III. Blockchain-Based Transaction Method

A. Direct Power-Purchase Approach

The blockchain transaction framework is presented in Fig. 2, which comprises three layers, including a business layer, a service layer, and an information layer that is on the basis of a federated chain platform. The three layers are interconnected and formed together as a full distributed transaction system. In the information layer, each block in the blockchain consists of a block header and a block body. The submitted transaction information during each trading process is documented in the block by the chain code, which supports the data operation for the service layer and business layer. In the service layer, market entities present their registration information by their own node types, and the certificate authority (CA) node of the corresponding type of institution authenticates the registration information for access. The authenticated market entity information is stored uniformly into the database for management, and the unique identification string of the system is generated according to the chain code and encryption algorithm. After that,

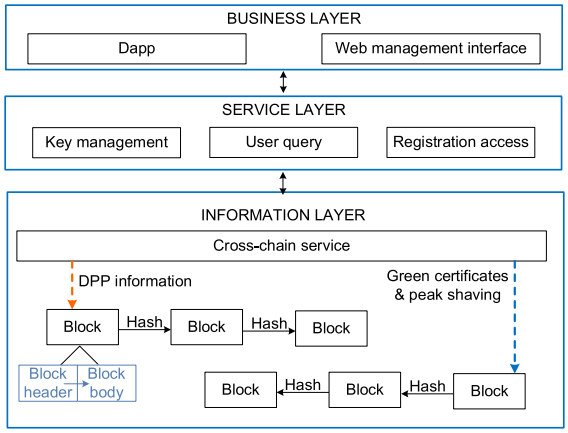


Fig. 2. Blockchain transaction framework.

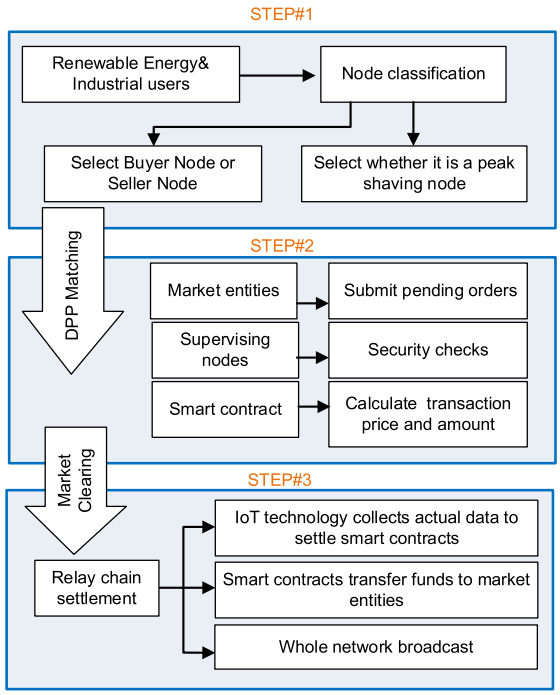


Fig. 3. Framework of DPP.

the public key and private key will be distributed. The public key is made publicly across the whole network, while the private key is only returned to the master node for local storage. The market entity that obtains the private key can obtain the trading authority of the trading application system. The business layer mainly consists of the distributed application system and the system front-end, including the interface of the regulator and the trading institution. It provides data exchange, information interaction and display functions during the market transaction and settlement processes.

The framework of DPP is shown in Fig. 3, which consists of three main steps. Each market entity first invokes the registration contract, pays a certain deposit, completes the access registration, and divides different types of nodes according to

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the identity of the subject. For the allocation of peak-shaving subsidies, the subjects of peak-shaving tasks are first determined. First, all thermal power units are involved in the peak shaving task, which is not classified here; for renewable energy units, some of the wind and solar photovoltaic units participate in load peak shaving, while the rest of the renewable energy units don't. Peak-shaving subsidies are granted to the units that participate in peak-shaving, including thermal units and renewable energy units. They are listed as priority nodes in the quota system and have the appropriate priority depending on their peak-shaving capacity. After entry, the DPP for each unit is completed through direct purchase matching contracts during the direct purchase cycle. Finally, the transaction results are verified and the DPP market is cleared. So far, the current round of DPP has been completed.

B. Quota System Between Power Sellers and Buyers

Market entities in the traditional DPP mainly include industrial users, power generation companies and trading institutions. Industrial users and power generators that comply with the access mechanism directly sign electricity purchase contracts in accordance with the previously agreed transaction methods. This approach breaks the monopoly market structure dominated by power companies. On the one hand, it can effectively reduce the power purchase cost of enterprises. On the other hand, it can be combined with renewable energy consumption policies to guide market development. The DPP model can be divided into two categories, namely, DPP for industrial users in the province and DPP across provinces according to the region. According to the transaction method, it can be divided into negotiated bilateral transactions, centralized matching transactions, centralized bidding transactions and centralized listing transactions. The trading center is responsible for publishing transaction content while the companies and power generators submit transaction requirements within a specified time.

The transaction center determines the two market entities (i.e., the generation companies and the industrial users) to the transaction according to a certain transaction method, and the dispatch center conducts security checks and congestion management on the matched transaction and obtains the restricted transaction result. The dispatch center finally conducts electric energy transmission in accordance with the signed contract. The centralized matching model is the most commonly used DPP, and the DPP transaction process is shown in Fig. 4.

The objective function of participating in the centralized matching transaction mode is maximizing the social welfare value, which is expressed as follows:

$$W = \max \left(\sum_{i=1}^{m} \sum_{j=1}^{n} P_{B,j} - P_{S,i} \right) Q_{ij} \tag{1}$$

The model includes constraints on the purchase of electricity $Q_{B,j}$ and constraints on the sales of electricity $Q_{S,i}$, shown as (2) and (3), respectively. Equation (2) represents that the amount of the electricity purchased from all sellers for buyer *j* is supposed to be no more than its demand. Similarly, (3) represents that the amount of the electricity sold to all buyers for seller *i* is supposed to be no more than its supply capability.

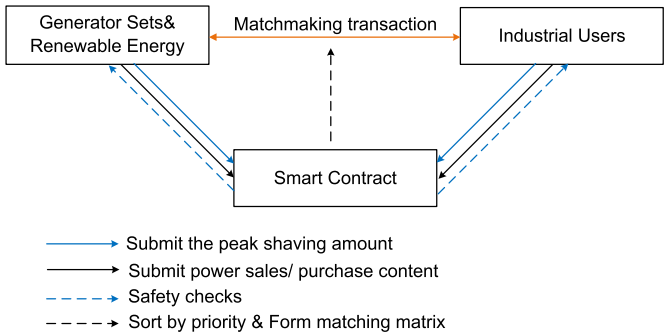


Fig. 4. Process of DPP.

$$\sum_{i=1}^{n} Q_{ij} \le Q_{B,j}, j = 1, 2, \ldots, m \tag{2}$$

$$\sum_{j=1}^{m} Q_{ij} \le Q_{S,i}, i = 1, 2, \ldots, n \tag{3}$$

A typical DPP problem solution methodology is the matching matrix method based on the social welfare of each transaction pair (i.e., the seller and buyer in the DPP transaction) [28], [29], which can determine the transaction order and the transaction quantity. The matching matrix is an $m \times n$ order matrix, which contains m buyers and n sellers. As shown in (4) and (5), the matrix element M_{ij} is composed of the price difference between the transaction pair and the transaction quantity, which represents the social welfare of each transaction pair between the seller and buyer.

$$[M_{ij}]_{m \times n} \tag{4}$$

$$M_{ij} = (P_{B,j} - P_{S,i} - S_{ij}) Q_{ij} \tag{5}$$

The transaction pair with highest M_{ij} in the matching matrix $[M_{ij}]_{m \times n}$ is preferentially matched. Thus, according to the objective function of the DPP problem shown in (1), the matching matrix directly determines the serial numbers of each round of transactions pair in descending order according to the M_{ij}. After confirming the transaction pair, the (2) and (3) are supposed to be checked. If one of constraints in the (2) is not hold (e.g., the electricity purchased from all sellers for buyer *j* is greater than its demand), the row *j* of the matching matrix will be removed. Once one of constraints in the (3) is not hold (e.g., the amount of the electricity sold to all buyers for seller *i* is greater than its supply capability), the column *i* of the matching matrix will be removed. After that, the matching matrix is update for the next round of matching until the social welfare is less than 0 and the transaction ends. In each round of matching, the settlement prices of the electricity purchaser and electricity seller are defined by (6) and (7) respectively [2].

$$P_j^i = P_{B,j} - \frac{\Delta F_{ij}}{\sigma} \tag{6}$$

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$$P_i^j = P_{S,i} + \frac{\Delta F}{\sigma} \tag{7}$$

In the formula, ΔF is defined by (8):

$$\Delta F = P_{B,j}(1-\Delta\beta) - \left(\frac{P_{S,i}}{1-\Delta\beta} + \tau + \Delta W\right) \tag{8}$$

The quota mechanism is a trading method by specifying the proportion of renewable energy consumption. Before the transaction starts, the government determines the proportion of renewable energy should be purchased by each industrial user, and prioritizes renewable energy to participate in the matching. For each pair of the seller and the buyer in the transaction (i.e., the renewable energy generation company and the industrial user), it is matched based on the element in the matching matrix $[M_{ij}]_{m\times n}$. After industrial users purchase a corresponding proportion of renewable energy, the remaining power demand is matched with thermal power.

To deal with the problem of renewable energy consumption and promote the development of renewable energy, electricity market launches a green certificate trading mechanism. For renewable energy power generation companies, after they generate a certain amount of green electricity, they will obtain a certain number of green certificates. And if the industrial users purchased the renewable energy power, they will also obatain the green certificates. The trading center allocates a certain amount of green energy quota to market entities based on actual conditions and needs, and requires market entities to return the green certificates after the end of the trading cycle to complete the quota. In the trading market, both entities to the transaction can purchase or sell green certificates according to their needs to complete the quota.

After considering the green certificates, the matching matrix element M_{ij} is expressed as (9).

$$M_{ij} = (P_{B,j} - P_{S,i} - S_{ij} - N_{gr}\cdot P_{gr})Q_{ij} \tag{9}$$

The settlement price of the electricity seller after considering the green certificates is expressed as (10).

$$P_i^j = P_{S,i} + \frac{\Delta F}{\sigma} + N_{gr}\cdot P_{gr} \tag{10}$$

From the above two equations, it can be seen that adding green certificates to the matching matrix can improve the priority of renewable energy transactions, and can bring more profits to renewable energy in the settlement process.

C. Direct Power-Purchase Market Mechanism Considering Peak Shaving Capacity of Industrial Users

Due to the fixed amount of power generation and transmission, the dispatching agency must adjust the operation mode of the load when the peak-to-valley difference is large. The peak-to-valley difference of the system is maintained within a reasonable range by adjusting the load.

Based on the impact of peak shaving on users, this paper uses peak shaving capability as one of the indicators for user transaction ranking. On the basis of the quota system, the transaction priority is set from high to low according to the load peak shaving capacity, and the users with more emphasis on the peak level are given priority to purchase electricity. This method ensures that the power system has sufficient peak shaving capacity, enhances the enthusiasm of units to participate in peak shaving and improves the consumption level of renewable energy through the quota system.

When conducting transaction matching, the quota ratio for purchasing renewable energy is first set, and electricity from renewable energy is preferentially purchased. The user side sorts according to the peak shaving capability and determines the priority of purchasing electricity. The peak shaving capacity is mainly determined by the comprehensive adjustable load, including start-stop peak reduction, shift peak reduction and incremental load peak reduction. The comprehensive adjustable load of industrial users is stated as:

$$L_x = w_\alpha^C \cdot L_\alpha^C + w_\beta^C \cdot L_\beta^C + w_\gamma^C \cdot L_\gamma^C \tag{11}$$

After the industrial users are ranked from high to bottom according to the peak shaving order, the renewable energy electricity sellers are sorted according to the electricity price from low to high, and the buyers and sellers trade according to the order. The thermal power seller completes the transaction of the remaining power purchase demand according to the same sorting method.

IV. Case Study

A. Simulation Setup Conditions

In terms of platform selection, this paper uses Z-ledger based on the Hyperledger Fabric core as the deployment platform [30]. Compared with the traditional trading platform of Ethereum, its authority management function can well meet the market access requirements. In addition, the trading platform supports custom smart contracts to facilitate the design of corresponding functions for different interaction requirements.

In the simulation hardware environment, this chapter simulates various electricity market entities on multiple computers, builds a peak shaving coupled DPP model, deploys the contract in the chain code container, and each entity implements the corresponding function by calling the contract. In addition, the DPP in the green certificate model is simulated on these computers, that is, the transaction entity completes the transaction process on the computer of the transaction institution.

The simulation is divided into four steps, which are listed as follows:

1) A computer simulation transaction node based on the Hyperledger Fabric platform is used to initiate DPP or peak shaving service requests. When a user or power generation company has an electric energy transaction, it executes mining and packs it into blocks;
2) Generate public and private keys and certificates. The certificate service of the Hyperledger platform simulates a regulatory agency in the DPP. If an enterprise wants to join the DPP system, it must submit an application to the certificate issuing authority for registration and obtain a signature and public and private keys.
3) Generate a Merkel tree. Each node regards the genesis block as the first block of the blockchain, that is, sets the

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TABLE I
CROSS-REGIONAL TRANSACTION LOSS AND TRANSMISSION COSTS

Area	Network loss rate(%)	Transmission cost(Token)
A-A	2	0.2
B-B	2	0.2
C-C	2	0.2
D-D	2	0.2
A-B	5	0.7
A-C	6	0.9
A-D	5.5	0.8

TABLE II
PURCHASE ORDERS OF LARGE USERS

User	Purchase quantity /MWh	Quoted price/(Token/kWh)
U_1	10095	5.365
U_2	19780	5.228
U_3	28795	5.453
U_4	18000	5.603

TABLE III
GENERATOR UNITS SALES ORDER

Thermal power companies	Declaration amount/MW·h	Quoted price/(Token/kWh)	environmental protection prices /(Token/kWh)
G_1	15017	3.378	0.15
G_2	20513	3.407	0.1
G_3	30093	3.249	0.1
G_4	7681	3.535	0.2

genesis block as the root node of the Merkel tree for data storage, thereby ensuring the safety and credibility of data related to DPP.

4) Start and run smart contracts. The required smart contracts mainly include smart contracts for DPP access, smart contracts for security checks by the supervision center, smart contracts for emission reduction rewards and punishments, smart contracts for transaction matching.

B. Simulation Assumptions

To analyze the DPP under different market mechanisms, the article starts with three trading schemes. The simulated DPP data is shown in the figure and table, and the network loss rate and transmission cost of the inter-regional power transmission are shown in Table I. Token refers to the transaction currency of the blockchain platform, set 1Token as 0.1 ¥.

Suppose that there are four industrial users U_1, U_2, U_3, U_4, of which U_1 and U_2 are in area B, U_3 is in area C, and U_4 is in area D. Each industrial user submits a pending order as shown in Table II, including the purchase of electricity and quotation of the pending order.

Assuming that there are 4 thermal power generators and 4 wind power generators participating in market transactions. The order of power generators status is shown in Table III.

The orders for renewable energy units are shown in Table IV.

According to the start-stop load, transfer load and increasable load declared by industrial users, the priority order of industrial users U_1 to U_4 is determined as shown in Table V. The weights of these three loads are respectively set to 0.5, 0.3, 0.2.

TABLE IV
RENEWABLE ENERGY SALES ORDER

Wind power companies	Declaration amount/MWh	Quoted price/(Token/kWh)
W_1	5950	3.600
W_2	6810	3.588
W_3	2457	3.490
W_4	6712	3.596

TABLE V
PEAK SHAVING ABILITY FOR INDUSTRIAL USER

	Start-stop load load/MWh	Shift peak shaving /MWh	Increasable load /MWh	Comprehensive regulation load /MWh	Prioritization
U_1	3.4	0.8	0.5	2.04	4
U_2	4.6	0	0.3	2.36	2
U_3	7.5	4.3	0	5.04	1
U_4	1.9	0	5.7	4.75	2

In this paper, we design the following four cases for comparison of the designed trading mechanism.

- Case A: Typical DPP market model. Wind power and thermal power jointly participate in electricity sales and bidding together;
- Case B: The model of wind power and thermal power participating in the DPP market according to the quota. In this case, the simulation platform simulates the initiation of electronic contracts, and trades with industrial users according to the 1:3 quota ratio of wind power and thermal power. After the wind power is cleared, the thermal power will complete the remaining electricity purchase by industrial users, and the final transaction process will be recorded by the blockchain platform;
- Case C: A DPP mode with green certificate market. This scheme initiates smart contracts on the blockchain through the simulation platform, and the chain code is recorded on a single chain. Assume that the thermal power unit must pay 0.02 green certificate for every 1MWh of thermal power produced. For every 1MWh of electricity produced by a wind turbine, 0.1 green certificate can be obtained. The prices of green certificates are based on four criteria of 1 Token, 1.5 Tokens, 2 Tokens and 2.5 Tokens. After the wind power generator completes the matching transaction, the remaining electric energy can be converted into green certificates and circulated in the market to obtain additional benefits;
- Case D: Considering the peak shaving capability of industrial users, the transaction priority order of industrial users is determined according to the start-stop load, shifting load, and increase load declared by industrial users. If the comprehensive adjustable loads are the same, the industrial users will be sorted according to the power purchase. According to the peak shaving sequence, firstly, the renewable energy is matched with industrial users, and the wind power purchase volume is determined according to the 1:3 quota of wind power and thermal power, and then

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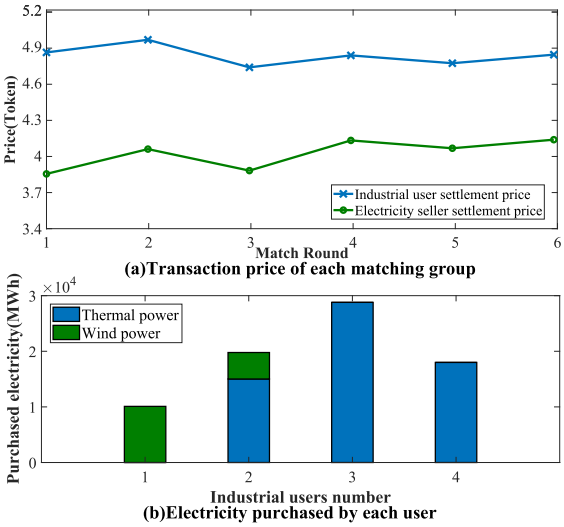


Fig. 5. Purchased electricity and price in Case A.

the surplus power purchased by industrial users is matched with thermal power units.

C. Results and Analysis

The purchased electricity and transaction price of Case A and Case B are shown as Figs. 5 and 6, respectively. Compared with Case A, the pending orders of the four wind power generation companies are all sold in Case B. Moreover, it can be found that the Case B increases the transaction volume of renewable energy and the consumption of renewable energy.

For Cace C, the total revenue of the four wind power companies W1-W4 at different green certificate prices are shown in Fig. 7. According to the result, each benefit contains both matching revenue and green certificate extra revenue components. In the Fig. 7, the matching revenue refers to the revenue obtained from the transaction between wind power enterprises and users. The additional revenue refers to the revenue obtained from the sale of green certificates for wind power that has not been sold through matching.

According to the result, when the price of green certificates is low, although W3 companies fail to sell, their remaining wind power can still be converted into green certificates to obtain revenue. As the price of green certificates increases, each wind power company has a higher priority in matching, and the revenue also increases.

In addition, different types of sellers have different revenue changes when the green certificate price changes. As shown in Fig. 8, with the increase of the price of green certificates, the overall revenue of wind power companies also increases. On the contrary, the revenue of thermal power companies has declined. This shows that the introduction of green certificates can bring huge profits to renewable energy.

The settlement prices before and after the introduction of green certificates are shown in Fig. 9. According to the result, the introduction of green certificates has increased the settlement price of wind power companies, enabling wind power companies

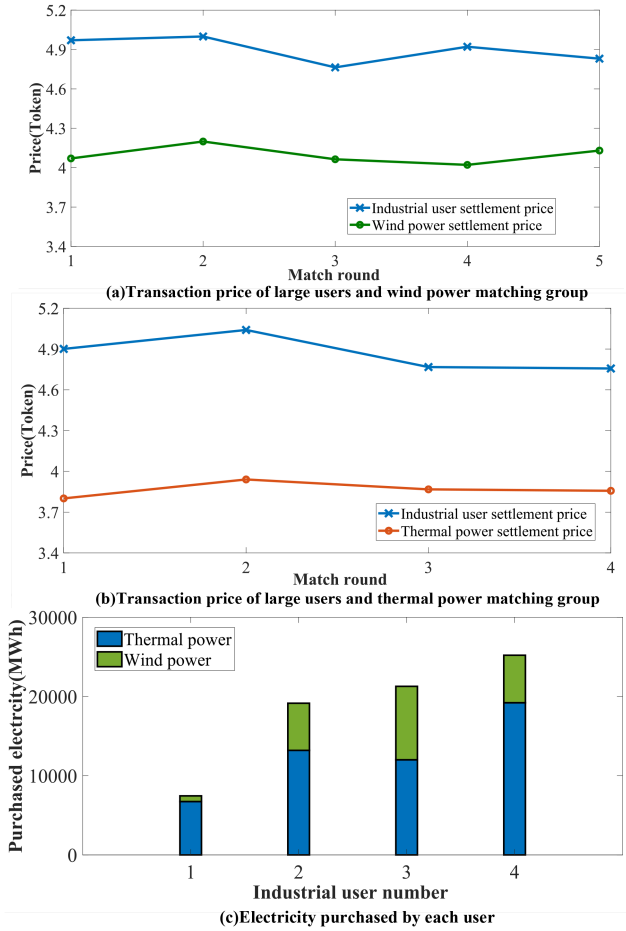


Fig. 6. Purchased electricity and transaction price in Case B.

to obtain more profits. On the contrary, as the settlement price of thermal power enterprises decreases, their profits are also decreased.

The simulation results of Case D are shown in Fig. 10. According to the results, the matching rounds in the transaction are sorted according to the given user's peak shaving ability. All industrial users first complete transactions with wind power companies, ensuring the consumption level of wind power.

D. Disussion

Many factors may affect the trading of renewable energy, here we discuss the impacts of different cases on the consumption and profit of renewable energy.

1) Renewable Energy Consumption: Promoting the consumption of renewable energy is an important issue in the current electricity market development. In this paper, we mainly consider wind power generation in renewable energy, and design four transaction methods: typical DPP, DPP based on quota system, DPP based on green certificate and DPP based on peak shaving. The electricity sold of wind power in the four cases are shown in Fig. 11.

According to the result, typical DPP has the lowest consumption level of wind power. Case B and Case D with quota system

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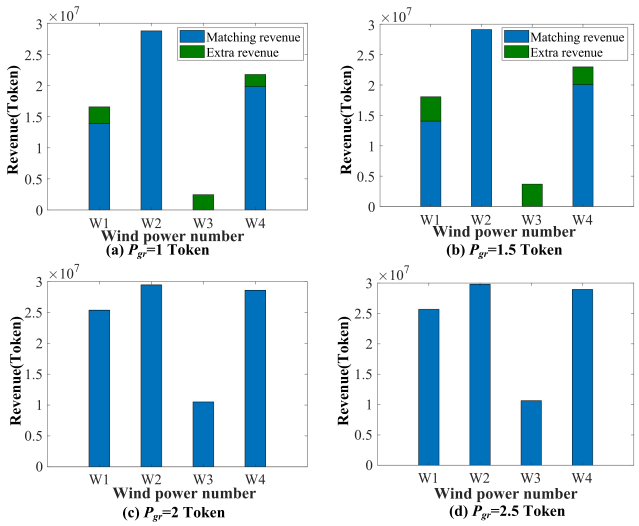


Fig. 7. Total revenue of wind power.

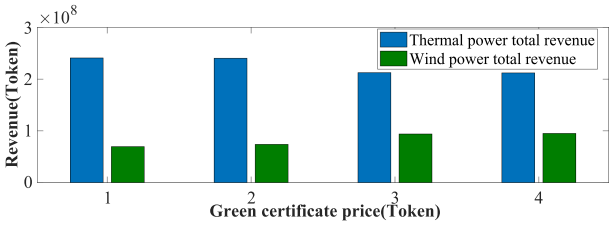


Fig. 8. Total revenue of thermal power and wind power.

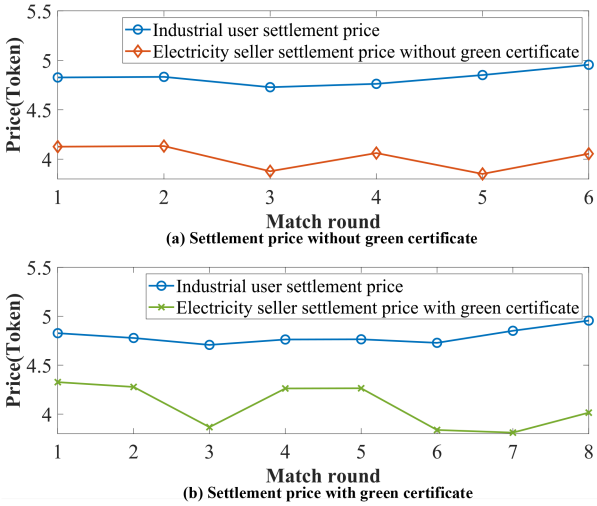


Fig. 9. The settlement price in Case C.

have the highest consumption level of wind power. Wind power consumption in Case C is determined by the green certificate price, and a higher green certificate price can bring more wind power consumption.

Case B and Case D both adopt the quota sales of renewable energy, the purchase ratio of wind power and thermal power is 1:3, and priority is given to arranging wind power for trading.

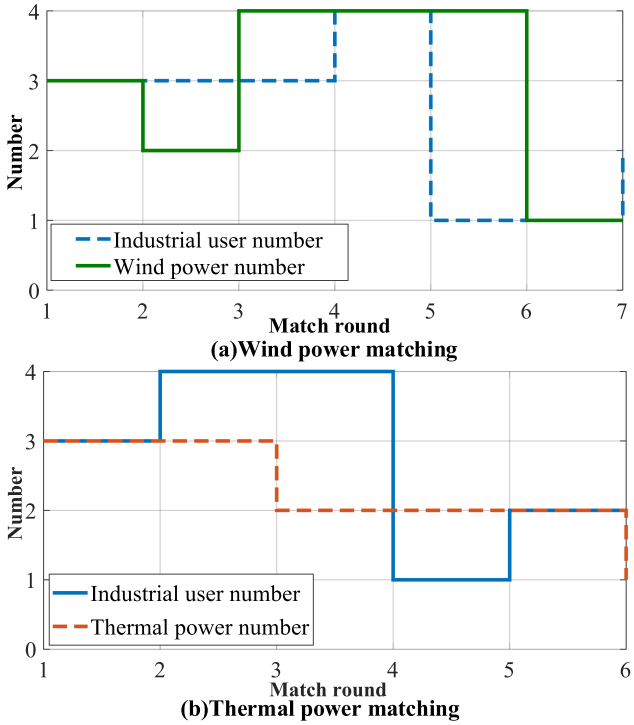


Fig. 10. Buyer and Seller Number.

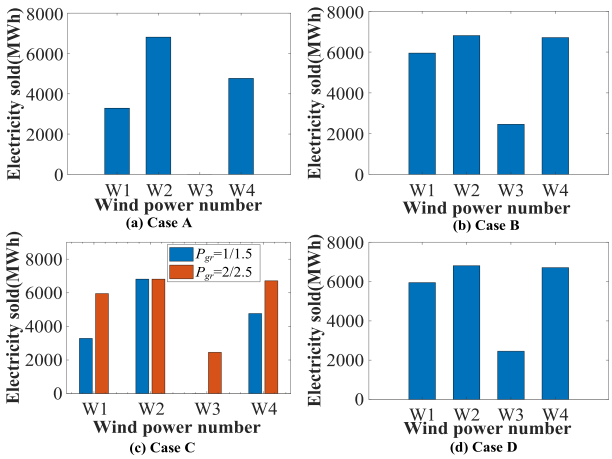


Fig. 11. Wind power trading capabilities.

This mechanism guarantees mandatory consumption of wind power. Compared with the results of typical DPP, the consumption of each wind power company has increased, indicating that the way of wind power quota promotes the consumption of wind power.

Since the green certificates are considered in the matching and settlement processes, the wind power consumption level of case C needs to be analyzed in combination with the green certificate price. Compared with the results of typical DPP, it can be seen that when the price of green certificates is low, the consumption level of wind power does not change; when the price of green certificates increases, wind power can get some

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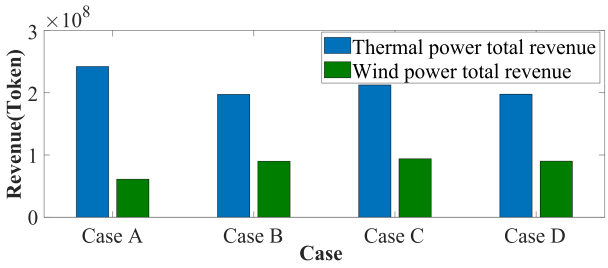


Fig. 12. Revenue in different cases.

priority in the matching process, so the wind power consumption level increases.

2) Total Revenue of Sellers: In addition to discussing the consumption levels in different cases, we also need to further analyze the profit of wind power. The wind power revenue of the four cases is shown in Fig. 12.

According to the result, compared with typical DPP, the other three trading mechanisms have higher wind power revenue. For Case B and Case D, the consumption level is the highest, and therefore the revenue is also the highest. For Case C, it has considered the green certificates as an additional means to boost renewable energy revenue. These three approaches promote the consumption of renewable energy with different focuses.

Compared with the centrally organized DPP, this blockchain based model can effectively avoid malicious competition. Moreover, distributed data storage can enhance transaction security. In addition, the platform construction based on the enterprise-level alliance chain can also prevent data from being tampered with and viewed by non-subject users.

V. Conclusion

Based on the characteristics of DPP, this paper designs different DPP transaction mechanisms in the electricity transaction based on blockchain, discusses the quota system, green certificate and peak regulation respectively, and simulates the transaction on the blockchain encryption test platform. In addition, renewable energy consumption and the influencing factors of revenue are discussed. The main findings of this paper are as follows: (1) The introduction of quota system can ensure the effective utilization of renewable energy. (2) The introduction of green certificates can bring more profits to renewable energy and help it better participate in market competition. (3) The introduction of peak shaving weight can reasonably utilize its peak shaving capacity while satisfying the consumption of renewable energy. (4) Transactions based on blockchain platforms can provide higher security for such large-value transactions. The load pattern [31], [32], [33] and power flow [34] will be greatly changed by different demand response programs [35] with the development of distribuited solar PV systems [36], [37] and the intergration of electricity vehicle [38], [39], [40] and solar PVs [41], [42], [43], [44], which cause much more uncertainty. Therefore, the future works of this paper are listed as follows:

1) Considering the impacts of the uncertainty of renewables energy on the proposed model.
2) With the rise of carbon trading in the electricity market, we need to consider introducing such methods in DPP transaction.
3) Considering the iterative upgrade of blockchain technology, we need to consider improving the platform used.

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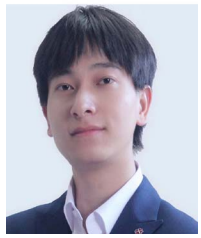
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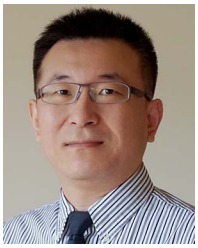


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